

CUSHING, OK, USA

WMO: 720357

Lat: 35.950N

Lon: 96.773W

Elev: 916

StdP: 14.22

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 53993

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 13.6	(c) 18.7	(d) 3.8	(e) 6.9	(f) 17.3	(g) 8.9	(h) 9.0	(i) 23.2	(j) 25.4	(k) 48.5	(l) 23.3	(m) 49.4	(n) 5.4	(o) 20	(p) 0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	100.4	76.2	97.4	76.1	94.8	76.4	80.2	93.0	78.9	91.8	77.8	90.4	8.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	146.1	86.8	75.3	137.4	85.1	73.5	128.9	83.0	44.6	93.0	43.0	92.2	41.8	90.2	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	19.3	17.2		4.8	104.7	8.1	5.0	-1.0	108.3	-5.8	111.2	-10.3	114.0	-16.3	117.6
			DB	3.7	81.4	7.6	1.8	-1.7	82.7	-6.2	83.8	-10.4	84.8	-15.9	86.1
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.8	39.0	42.0	52.5	60.1	68.8	78.2	81.9	81.3	73.4	61.3	50.4	40.1		
	DBStd	17.65	10.46	11.56	10.93	9.46	7.82	5.00	5.24	5.83	6.69	9.27	9.45	9.68		
	HDD50	1213	364	272	99	23	2	0	0	0	0	16	108	329		
	HDD65	3528	807	647	408	201	54	0	0	0	11	183	446	771		
	CDD50	5171	22	47	176	326	584	847	989	971	701	366	119	23		
	CDD65	2010	0	2	20	53	171	398	525	506	261	68	6	0		
	CDH74	22429	4	25	182	556	1594	4350	6401	5912	2635	696	72	2		
	CDH80	10761	0	4	40	138	536	2021	3484	3199	1135	197	7	0		
(14) Wind	WSAvg	6.7	7.1	7.6	8.2	8.7	7.0	6.6	5.3	5.0	5.0	6.2	7.0	6.7		
(15) Precipitation	PrecAvg	37.5	1.3	1.5	3.0	3.8	5.0	5.2	3.2	2.9	3.9	3.3	2.5	1.9		
	PrecMax	52.8	3.9	4.3	8.5	8.1	11.0	12.8	7.5	7.4	7.2	10.0	6.8	5.1		
	PrecMin	28.6	0.0	0.0	0.5	0.3	1.3	1.6	0.8	0.1	0.9	1.0	0.0	0.1		
	PrecStd	5.7	1.0	1.2	1.9	2.0	2.7	2.9	1.9	1.8	1.6	2.4	2.1	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.1	77.5	85.5	89.8	91.3	99.4	106.3	108.5	99.7	89.6	81.1	72.2		
		MCWB	55.5	60.1	64.4	68.0	72.2	77.3	76.8	76.3	75.4	70.6	63.6	61.2		
	2%	DB	67.7	71.3	79.2	83.6	88.6	95.3	100.4	102.0	93.2	85.8	75.1	65.3		
		MCWB	54.7	56.6	61.6	66.6	71.8	76.7	76.3	75.7	75.1	70.2	61.5	57.2		
	5%	DB	61.3	65.8	73.4	80.7	86.2	91.7	97.4	98.2	90.2	81.5	71.4	60.9		
		MCWB	50.5	54.8	59.8	64.4	71.2	75.4	76.2	76.2	73.9	66.8	59.2	51.7		
	10%	DB	55.4	61.1	70.2	76.9	82.2	90.2	94.8	94.6	86.7	77.5	66.2	55.3		
		MCWB	45.8	52.3	58.5	63.0	69.9	75.2	76.7	76.6	72.5	65.3	57.1	48.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.0	62.8	67.8	71.7	77.0	80.7	82.5	81.8	79.4	75.0	67.9	65.3		
		MCDB	67.9	71.5	78.8	80.9	86.7	93.2	94.3	95.2	94.5	84.9	74.9	72.5		
	2%	WB	55.5	59.8	65.4	69.3	74.6	79.3	80.3	79.7	77.3	72.9	64.2	58.8		
		MCDB	63.6	68.7	75.0	78.0	84.1	91.1	92.5	94.0	90.0	81.7	70.6	62.8		
	5%	WB	51.2	56.8	62.9	67.6	73.1	78.1	78.9	78.4	75.7	70.9	61.9	53.7		
		MCDB	59.4	63.7	72.0	76.2	82.2	89.3	91.6	92.0	86.2	77.2	68.2	58.3		
	10%	WB	47.1	52.1	60.2	65.7	71.7	76.8	77.7	77.3	74.1	68.2	58.7	49.6		
		MCDB	55.5	61.3	68.3	74.2	80.3	87.4	90.5	90.2	83.5	74.4	65.3	56.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.8	21.7	23.2	23.2	20.5	20.2	21.7	21.7	22.7	23.7	23.6	20.1		
		MCDBR	31.5	29.5	27.6	27.2	23.7	23.1	25.5	26.8	26.2	26.6	27.9	27.9		
	5% WB	MCWBR	19.2	17.5	14.9	14.1	10.9	8.9	7.9	7.9	10.7	13.3	16.2	18.6		
		MCWBR	28.8	25.8	22.8	23.4	19.8	20.1	21.8	22.2	22.0	20.9	24.7	23.2		
(40) Clear-Sky Solar Irradiance	taub		0.297	0.303	0.342	0.381	0.415	0.430	0.434	0.433	0.391	0.348	0.316	0.292		
		taud	2.504	2.502	2.419	2.325	2.298	2.291	2.320	2.264	2.371	2.479	2.508	2.528		
	Ebn at Noon		288	299	294	286	276	271	269	267	274	279	278	283		
		Edh at Noon	26	29	35	40	42	42	41	42	36	30	25	24		
(44) All-Sky Solar Radiation	RadAvg		844	1036	1354	1686	1842	2111	2125	1923	1597	1204	893	713		
	RadStd		85	157	84	116	185	200	116	121	139	149	113	81		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page